

Aditya Rahalkar

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EDUCATION

Virginia Tech

Doctor of Philosophy (Ph.D.) in Mechanical Engineering — GPA: 3.95/4

Blacksburg, VA

Aug 2024 – Present

College of Engineering Pune

Bachelor of Technology in Mechanical Engineering — GPA: 8.71/10 | Rank: 7/166

Pune, India

Aug 2020 – May 2024

EXPERIENCE

Graduate Research Assistant

CVeSS, Virginia Tech

Blacksburg, VA

Aug 2024 – Present

Rail Lubricity Sensing & Analysis (FRA-sponsored)

- Designed and developed a laser-based optical sensing system to detect and quantify flange grease presence on rail tracks, advancing the system from an initial proof-of-concept to a real-time detection platform capable of performing lubrication sensing, data analysis, and visualization simultaneously in the field.
- Built a rail lubricity data analysis and visualization application enabling real-time and historical evaluation of track condition and lubrication performance trends.
- Developed a machine learning model to predict the Thickness Index of grease present on the rail, supporting data-driven maintenance decisions for track engineers.

Bayesian Inspection Framework (FRA-sponsored)

- Contributing to a Bayesian modeling initiative to build a fact-based framework combining automated (ATGMS) and manual track inspection data.
- Applying statistical and Bayesian analysis methods to evaluate the relationship between inspection type/frequency and derailment risk, supporting FRA rulemaking and helping U.S. Class I railroads plan maintenance and inspection schedules more effectively.

General Engineer

Volpe National Transportation Systems Center, US DOT (Headquarters)

Washington, DC

June 2026 – Present

- Applied data analysis and visualization techniques to translate complex railroad testing data into clear, actionable insights for track maintenance personnel.
- Analyzed wear data collected from special trackwork structures, such as diamond crossings, and developed visualizations to characterize wear patterns and support maintenance prioritization.
- Developed a user interface for visualizing track stability and wheel-climb derailment data, supporting engineers in assessing derailment risk across the rail network.

Computer Vision Intern

LightRay Technologies

Pune, India

September 2023 – May 2024

- Designed and developed a rear-end critical object detection system for two-wheelers, aimed at improving rider safety by identifying hazards approaching from behind.
- Employed the YOLOv8 Nano framework to build a lightweight, real-time object detection and classification pipeline suited for deployment on resource-constrained hardware.
- Incorporated distance, speed, and direction parameters into the detection pipeline to robustly identify critical objects and generate timely alerts for the rider.

- Analyzed real-time cutting-force signals collected from milling experiments and applied decision-tree-based feature selection to identify the most informative signal features for tool-condition classification.
- Developed a KNN-based white-box model, with tuned hyperparameters, for increased transparency and interpretability in the classification of tool condition.
- Achieved 98% accuracy in tool condition classification by eliminating Type II errors through data pre-processing, including noise reduction and outlier detection.

PUBLICATIONS

1. **Rahalkar, A.**, S. M. H. Mirzaei, Y. Chen, C. Holton, and M. Ahmadian. “Laser-Induced Fluorescence Sensing of Flange Lubrication: Track Cart Testing.” *International Journal of Rail Transportation*, 2026.
2. R. M. Wahul, **A. Rahalkar**, O. Khare, R. N. Soman, and A. Patange. “Making Informed Decisions in Cutting Tool Maintenance in Milling: A KNN-Based Model Agnostic Approach.” *Eksploracja i Niezawodność – Maintenance and Reliability*, 28(3), 2026.
3. **Rahalkar, A.**, S. M. Mirzaei, Y. Chen, C. Holton, and M. Ahmadian. “Evaluation of Flange Grease on Revenue Service Tracks Using Laser-Based Systems and Machine Learning.” *Infrastructures*, 10(4), 80, MDPI, 2025.
4. A. Sanju, A. D. Patange, **A. M. Rahalkar**, and R. Soman. “Notifying Type-2 Error and Segregating Undefined Conditions in Health Monitoring of Milling Cutter: A Statistical and Deep Learning Approach.” *Journal of Vibration Engineering & Technologies*, 13(1), 35, 2025.
5. O. M. Khare, S. Gandhi, **A. Rahalkar**, and S. Mane. “YOLOv8-Based Visual Detection of Road Hazards: Potholes, Sewer Covers, and Manholes.” *2023 IEEE Pune Section International Conference (PuneCon)*, 2023.
6. **Rahalkar, A.**, Mehdi Ahmadian, et al. “Laser-Based Systems for Onboard Detection of Flange Grease on Revenue Service Tracks.” Presented at the AREMA Annual Conference, 2025.
7. **Rahalkar, A.**, Mehdi Ahmadian, et al. “Where’s the Grease? Turning Rail Data into Easy Visuals for Better Maintenance.” Presented at the Big Data Conference, 2025.